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Please fill in answers in the space supplied and show all your work to receive full credit.
Useful Tables at the end of the Exam.

1. (21 points)

A) State whether the following statements are **True or False (3 pts each)**

_____ A pump does work on a gas to increase the pressure and/or elevation.

_____ *Transient* operation denotes a change in state with time.

_____ As velocity decreases in a diffuser, pressure decreases.

_____ For a closed system, entropy may be produced within the system.

_____ The maximum coefficient of performance of *any* refrigeration cycle operating between cold and hot reservoirs at 40°C and 80°C, respectively, is closely 8._____ Entropy production, σ , is path-dependent.

_____ The second Carnot corollary states that all power cycles operating between the same two thermal reservoirs have the same thermal efficiency.

B) (4 pts) An inventor suggests an improvement on the power plant cycle. Aware that one half of the heat transferred to the steam in the furnace is discarded in the condenser to the environment, the inventor suggests getting rid of this wasteful component and sending the steam to the pump as soon as it leaves the turbine (cycle contains boiler, turbine and pump). Do you think this new design will work? Explain.

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2. (35 pts) A heat pump is used to meet the heating requirements of a house and maintain it at 20°C in winter. On a day when the average outdoor temperature drops -2°C, the house is estimated to lose heat at a rate of 80,000 kJ/h. If the heat pump under these conditions has a COP of 2.5,
- a. (10 pts) Draw a schematic showing the heat pump cycle, the thermal reservoirs, T_H , T_C , $\dot{W}_{cycle}$, $\dot{Q}_C$ and $\dot{Q}_H$. Show the direction of the energy transfer rates with arrows.
 - b. (5 points) Determine the power consumed by the heat pump.
 - c. (5 points) Determine the rate at which heat is absorbed from the cold outdoor air.
 - d. (5 points) Determine the minimum power required to drive this heat pump.
 - e. (5 points) Determine the total heating cost per day (12 cents/ kWh of electricity) of using heat pump. If we were to use electric resistance heater instead of heat pump, how much the cost of heating will be?
 - f. (5 points) Determine the entropy production, σ and state if the cycle is possible, impossible or reversible. Support your answer for full credit.

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3. (40 points) An air turbine operating at steady state is providing power to both a pump and an air compressor. The mass flow rate of the air at the turbine inlet is 4.3 kg/s at 740K and 5 bar. The air exits the turbine at 380K and 1 bar. The turbine provides power at a rate of 850 kW to the compressor and the rate of heat transfer from the turbine to its surroundings is 4500 kJ/min. Air can be modeled as an ideal gas and kinetic and potential energy changes are negligible.

- a) (6 pts) Draw a schematic of the system and label the system boundary.
- b) (5 pts) State the engineering model
- c) (4 pts) Write the simplified energy rate balance
- d) (8 pts) Determine the volumetric flowrate of the air at the turbine exit, in m³/s.
- e) (10 pts) Determine the power to the pump in kW.
- f) (7 pts) Sketch the process from State 1 to State 2 on a P-v diagram.

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